

HESPERO AGCM, AB, CanadaWMO: **715400**Lat: **52.310N**Lon: **114.486W**Elev: **3166**StdP: **13.09**Time Zone: **-7.00 (NAM)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-24.3	-18.0	-30.1	1.1	-24.0	-24.0	1.6	-17.8	25.2	19.2	22.1	14.3	2.6	120	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	80.5	61.4	77.6	60.5	74.9	59.2	65.5	75.9	63.3	73.8	61.3	71.3	7.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.5	91.9	71.3	59.2	84.5	68.3	57.2	78.6	65.9	32.1	75.8	30.4	74.0	28.8	71.4	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	18.6	16.0	DB	-32.4	87.3	4.5	3.6	-35.7	89.9	-38.3	92.1	-40.8	94.1	-44.1	96.7	(4)
(5)			WB	-32.6	70.1	4.4	3.2	-35.8	72.4	-38.4	74.3	-40.9	76.1	-44.1	78.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	15.1	11.9	22.9	36.9	48.9	55.6	59.8	58.5	50.0	38.0	23.2	11.9		
	DBStd	20.41	13.77	14.23	13.54	8.80	7.40	5.08	4.43	5.09	7.57	8.33	12.97	13.37		
	HDD50	5969	1082	1068	840	396	111	12	0	4	90	378	805	1183		
	HDD65	10528	1547	1488	1305	842	500	283	168	206	451	836	1254	1648		
	CDD50	929	0	0	0	4	76	180	304	268	89	7	1	0		
	CDD65	13	0	0	0	0	0	1	7	5	0	0	0	0		
	CDH74	777	0	0	0	3	73	71	268	247	111	4	0	0		
	CDH80	111	0	0	0	0	5	5	39	44	18	0	0	0		
	WSAvg	6.5	5.8	5.5	6.3	8.0	8.2	7.5	6.7	5.9	6.0	6.5	5.7	5.4		
	PrecAvg	20.6	0.9	0.7	0.9	1.1	2.5	4.2	3.6	2.9	1.6	0.9	0.9	0.7		
	PrecMax	25.1	2.3	1.5	1.9	3.7	4.9	7.2	8.3	4.9	3.4	2.1	2.8	1.5		
(15) Precipitation	PrecMin	15.0	0.1	0.1	0.1	0.2	1.3	0.8	1.0	0.6	0.2	0.4	0.3	0.2		
	PrecStd	2.6	0.6	0.4	0.5	0.8	0.9	1.5	1.7	1.2	1.0	0.5	0.6	0.4		
	DB	48.7	46.0	55.5	72.7	80.4	79.2	83.8	84.8	82.8	71.7	59.4	46.7	37.5		
	MCWB	37.8	36.0	42.6	51.0	55.3	60.4	65.0	62.2	58.0	51.7	46.1	36.9	30.2		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	44.5	39.6	48.8	65.8	75.7	76.0	80.1	79.5	77.1	64.2	50.6	40.5	33.1		
	2% MCWB	36.4	32.8	38.6	47.8	53.3	58.9	64.8	62.2	57.6	48.4	40.3	33.1	29.8		
	5% DB	40.0	35.9	44.4	58.0	71.4	72.6	76.8	76.4	72.6	59.1	44.4	34.8	30.2		
	5% MCWB	33.8	31.1	36.4	43.0	52.0	57.0	62.7	61.6	56.4	46.3	37.0	29.8	26.7		
	10% DB	34.7	32.2	40.1	52.7	66.5	69.5	73.6	72.9	67.3	53.8	39.9	30.2	26.7		
	10% MCWB	30.4	28.7	34.2	41.0	49.7	55.5	61.2	60.3	54.0	43.2	34.4	26.7	23.9		
	0.4% WB	38.6	37.7	43.6	51.0	56.7	64.3	69.4	67.6	61.1	53.6	46.7	37.5	30.2		
	0.4% MCDB	46.7	44.2	54.1	71.3	75.6	74.8	78.2	76.4	76.4	69.1	56.8	45.3	33.6		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	36.7	33.8	39.6	47.7	54.4	61.1	66.5	64.8	58.9	49.9	41.1	33.6	29.8		
	2% MCDB	44.0	38.8	47.7	63.9	71.9	71.9	76.6	75.6	72.6	62.1	48.5	39.7	30.2		
	5% WB	34.1	31.4	36.8	44.3	52.7	58.8	64.3	62.8	56.9	46.9	37.8	30.3	26.9		
	5% MCDB	39.5	35.2	43.4	56.1	68.1	68.7	74.9	73.4	70.2	57.5	43.4	34.7	26.9		
	10% WB	30.5	28.9	34.4	41.4	50.8	57.0	62.1	60.8	54.5	44.2	34.7	26.9	23.9		
	10% MCDB	34.5	32.3	39.4	50.8	64.3	67.2	71.6	71.2	66.4	52.5	39.1	30.2	23.9		
	MDBR	21.1	22.1	21.4	21.6	25.0	22.6	23.5	24.9	25.9	23.9	20.5	19.7	19.0		
	5% DB	26.1	24.7	26.1	35.3	35.4	29.9	30.7	33.4	38.3	34.3	26.8	25.4	19.0		
(36) Mean Daily Temperature Range	MCWBR	18.5	18.6	17.1	18.9	16.2	15.1	17.2	18.2	19.9	19.8	17.6	19.0	19.0		
	5% WB	25.5	22.9	24.4	32.4	30.8	25.5	28.4	29.3	34.5	32.4	25.1	25.5	19.8		
	MCWBR	18.4	17.6	16.6	17.9	14.6	13.8	17.3	17.1	18.4	19.4	17.5	19.8	19.8		
	MCWBR	18.4	17.6	16.6	17.9	14.6	13.8	17.3	17.1	18.4	19.4	17.5	19.8	19.8		
(40) Clear-Sky Solar Irradiance	taub	0.257	0.261	0.300	0.331	0.344	0.351	0.349	0.360	0.313	0.282	0.253	0.243	0.243		
	taud	2.355	2.431	2.399	2.361	2.377	2.382	2.419	2.367	2.476	2.555	2.491	2.351	2.351		
	Ebn at Noon	242	279	285	287	288	286	284	274	278	268	247	228	228		
	Edh at Noon	18	23	29	35	36	36	34	34	27	20	16	15	15		
(44) All-Sky Solar Radiation	RadAvg	325	619	1054	1451	1710	1812	1940	1614	1134	686	363	251	251		
	RadStd	19	41	62	114	123	138	122	107	125	71	28	17	17		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page